Independent Component Analysis for Signal Separation

Theory, comparison with PCA, and a reproducible audio demonstration

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github.com/loukili27/ICA-Voices

Abstract

Signals encountered in audio, biomedical sensing, telecommunications, and finance are often mixtures of several latent phenomena. Independent Component Analysis (ICA) is a statistical framework for estimating latent components that are as independent as possible from multivariate observations. This report preserves the theoretical development of the original study and extends it with a direct comparison to Principal Component Analysis (PCA), mathematical derivations of whitening and FastICA, and a reproducible audio experiment. Three mono mixtures containing 264,515 samples each at 44.1 kHz are processed with FastICA. The algorithm converges in four iterations, yields linearly uncorrelated components with distinct higher-order statistics, and reconstructs the observations with a relative root mean square error of 4.219×10^{-16}. Because isolated reference sources are unavailable, these measurements validate numerical consistency rather than perfect source recovery. The complete notebook and audio files are available in the companion GitHub repository.

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1 Introduction

Signals are ubiquitous and frequently overlap. Voices recorded in a room, readings collected by multiple sensors, or electrical potentials measured on the scalp can each contain contributions from several underlying sources. Separating those sources is essential when the mixtures obscure the information of interest.

Blind source separation seeks to recover latent signals while knowing little or nothing about the sources or the physical mixing process. Independent Component Analysis (ICA) provides a principled solution when the observations can be modeled as linear mixtures of statistically independent, mainly non-Gaussian sources [1, 3]. Instead of relying only on covariance, ICA exploits higher-order and information-theoretic structure.

This distinction is important because Principal Component Analysis (PCA) and ICA are sometimes confused. PCA identifies orthogonal directions that preserve variance and is extremely useful for compression, visualization, and denoising. ICA asks a different question: which latent directions make the recovered signals as independent as possible? PCA may prepare the data through whitening, but independence is stronger than decorrelation.

This report develops the ICA model and its assumptions, derives the main mathematical criteria used for estimation, compares ICA with PCA, reviews representative applications, and concludes with a fully reproducible audio demonstration using FastICA.

2 Theoretical foundation of ICA

2.1 Linear instantaneous mixing model

Let the unknown source vector be

$$\mathbf{s}(t) = [s_1(t), s_2(t), \ldots, s_n(t)]^\mathsf{T}, \tag{1}$$

and let the observed mixtures be

$$\mathbf{x}(t) = [x_1(t), x_2(t), \ldots, x_m(t)]^\mathsf{T}. \tag{2}$$

The basic ICA model assumes an instantaneous linear mixture

$$\mathbf{x}(t) = \mathbf{A}\mathbf{s}(t), \tag{3}$$

where $\mathbf{A} \in \mathbb{R}^{m\times n}$ is an unknown mixing matrix. Each coefficient a_{ij} represents the contribution of source j to observation i.

In the determined setting, $m = n$ and $\mathbf{A}$ is invertible. ICA estimates an unmixing matrix $\mathbf{W}$ so that

$$\mathbf{y}(t) = \mathbf{W}\mathbf{x}(t) = \mathbf{W}\mathbf{A}\mathbf{s}(t) \approx \mathbf{s}(t). \tag{4}$$

$$\boxed{\mathbf{s}(t)} \xrightarrow{\mathbf{A}} \boxed{\mathbf{x}(t)} \xrightarrow{\mathbf{W}} \boxed{\mathbf{y}(t) \approx \mathbf{s}(t)}$$

independent sources — observed mixtures — estimated components

Figure 1: Linear instantaneous ICA model. The unknown matrix $\mathbf{A}$ mixes the latent sources, while ICA estimates an unmixing matrix $\mathbf{W}$ from the observations.

Figure 2 retains the intuitive illustration from the original report: several physical sources contribute simultaneously to each microphone signal, and ICA attempts to recover one latent component per output. It complements the compact matrix representation in Figure 1.

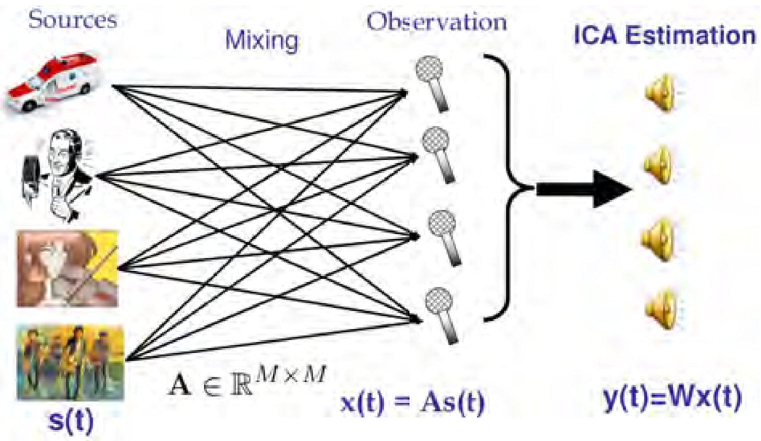

Figure 2: Intuitive blind source separation example retained from the original project materials. The source signals $\mathbf{s}(t)$ are mixed into microphone observations $\mathbf{x}(t) = \mathbf{A}\mathbf{s}(t)$, before ICA estimates $\mathbf{y}(t) = \mathbf{W}\mathbf{x}(t)$. The original external source of this legacy illustration should be confirmed before public redistribution.

2.2 Statistical independence

The central assumption is that the sources are mutually independent. For all admissible values,

$$p_{s_1,\ldots,s_n}(s_1,\ldots,s_n) = \prod_{i=1}^{n} p_{s_i}(s_i). \tag{5}$$

Independence implies zero covariance when moments exist, but the converse is generally false. Two variables can be uncorrelated while remaining dependent through nonlinear or higher-order relationships. This is precisely why diagonalizing a covariance matrix with PCA does not, in general, solve the ICA problem.

2.3 Non-Gaussianity and identifiability

The classical linear ICA model permits at most one Gaussian source. A multivariate Gaussian distribution that has been whitened is rotationally symmetric: any orthogonal rotation produces the same distribution. Therefore covariance and Gaussian likelihood alone cannot identify a unique rotation. Non-Gaussian source distributions break this rotational ambiguity and make the independent directions identifiable, apart from unavoidable permutation and scaling ambiguities [4].

The central-limit intuition is that a normalized sum of independent random variables tends to be more Gaussian than the original variables. ICA algorithms can therefore search for projections that are maximally non-Gaussian.

2.4 Intrinsic ambiguities

Even under ideal assumptions, ICA cannot determine the original component order because

$$\mathbf{A}\mathbf{s} = (\mathbf{A}\mathbf{P}^{-1})(\mathbf{P}\mathbf{s}) \tag{6}$$

for any permutation matrix $\mathbf{P}$. It also cannot determine component scale because a source may be multiplied by a nonzero scalar if the corresponding column of $\mathbf{A}$ is divided by the same scalar. Consequently, sign changes, scale changes, and component permutations do not indicate failure.

3 Mathematical criteria and estimation

3.1 Centering and PCA whitening

ICA is usually applied after centering and whitening. If $\boldsymbol{\mu} = \mathbb{E}[\mathbf{x}]$, centering gives

$$\mathbf{x}_c = \mathbf{x} - \boldsymbol{\mu}. \tag{7}$$

Let the covariance eigendecomposition be

$$\mathbf{C}_x = \mathbb{E}[\mathbf{x}_c \mathbf{x}_c^\top] = \mathbf{E}\mathbf{D}\mathbf{E}^\top. \tag{8}$$

A PCA whitening transform is

$$\mathbf{z} = \mathbf{D}^{-1/2}\mathbf{E}^\top \mathbf{x}_c, \qquad \mathbb{E}[\mathbf{z}\mathbf{z}^\top] = \mathbf{I}. \tag{9}$$

Whitening removes second-order correlations and constrains the remaining ICA mixing transform to a rotation. It simplifies the optimization but does not by itself establish independence.

3.2 Kurtosis

For a zero-mean variable y, the fourth-order cumulant is

$$\mathrm{kurt}(y) = \mathbb{E}[y^4] - 3\big(\mathbb{E}[y^2]\big)^2. \tag{10}$$

A Gaussian variable has zero excess kurtosis. Positive and negative values characterize super-Gaussian and sub-Gaussian behavior, respectively. Maximizing the absolute kurtosis of standardized projections is one possible ICA contrast, although kurtosis can be sensitive to outliers.

3.3 Negentropy

Differential entropy is

$$H(y) = -\int p_y(u) \log p_y(u)\, du. \tag{11}$$

Among variables with equal variance, the Gaussian distribution has maximum entropy. Negentropy therefore defines a nonnegative measure of non-Gaussianity:

$$J(y) = H(y_{\mathrm{Gauss}}) - H(y) \geq 0. \tag{12}$$

Because exact entropy estimation is difficult, FastICA uses robust approximations based on smooth nonlinear functions such as $G(u) = \log\cosh(u)$ [2].

3.4 Mutual information

Mutual information measures dependence between the components of $\mathbf{y}$:

$$I(y_1, \ldots, y_n) = \sum_{i=1}^{n} H(y_i) - H(\mathbf{y}) \geq 0. \tag{13}$$

It equals zero if and only if the components are mutually independent under standard regularity conditions. For whitened data, minimizing mutual information is closely connected to maximizing component non-Gaussianity [1].

3.5 Maximum likelihood

If $\mathbf{y} = \mathbf{W}\mathbf{x}$ and the source densities p_i are modeled, the change-of-variables formula gives

$$p_{\mathbf{x}}(\mathbf{x}; \mathbf{W}) = |\det \mathbf{W}| \prod_{i=1}^{n} p_i(\mathbf{w}_i^{\top}\mathbf{x}). \tag{14}$$

For T samples, the log-likelihood is

$$\ell(\mathbf{W}) = T \log |\det \mathbf{W}| + \sum_{t=1}^{T}\sum_{i=1}^{n} \log p_i(\mathbf{w}_i^{\top}\mathbf{x}(t)). \tag{15}$$

Maximizing this expression encourages an invertible transform whose outputs follow factorized source models.

3.6 FastICA fixed-point update

For whitened samples $\mathbf{z}$ and a unit vector $\mathbf{w}$, one FastICA update has the form

$$\mathbf{w}^{+} = \mathbb{E}\Big[\mathbf{z}\, g(\mathbf{w}^{\top}\mathbf{z})\Big] - \mathbb{E}\Big[g'(\mathbf{w}^{\top}\mathbf{z})\Big]\,\mathbf{w}, \qquad \mathbf{w} \leftarrow \frac{\mathbf{w}^{+}}{\|\mathbf{w}^{+}\|_2}. \tag{16}$$

The parallel algorithm updates all rows of $\mathbf{W}$ and then applies symmetric decorrelation,

$$\mathbf{W} \leftarrow (\mathbf{W}\mathbf{W}^{\top})^{-1/2}\mathbf{W}, \tag{17}$$

until the directions stabilize. The fixed-point approach is fast and avoids manually selecting a learning rate [2].

4 PCA versus ICA

PCA and ICA are related linear methods, but they answer different questions. PCA asks which orthogonal directions retain the greatest variance and orders them by explained variance [5]. ICA asks which latent directions make the recovered components as statistically independent as possible. The appropriate method therefore depends on the objective, not simply on which algorithm gives uncorrelated outputs.

Table 1: Conceptual comparison of PCA and ICA.

Criterion	PCA	ICA
Objective	Maximize explained variance	Minimize statistical dependence
Output relation	Orthogonal and uncorrelated	As independent as possible
Statistics	Second-order covariance	Higher-order and information-theoretic structure
Typical use	Compression, visualization, denoising, feature reduction	Blind source separation and artifact removal
Ordering	Ranked by explained variance	No intrinsic ranking
Ambiguities	Sign	Permutation, sign, and scale
Main assumptions	Finite covariance; orthogonal variance directions	Independent, mainly non-Gaussian latent sources and a suitable mixing model

When to use PCA. PCA is the natural choice when the goal is to reduce the number of variables, preserve as much variance as possible, visualize high-dimensional observations, remove low-variance noise, or construct compact features for a downstream model. Its ordered components make it easy to select a target dimension through the explained-variance ratio. Typical examples include exploratory data analysis, image compression, preprocessing before regression or clustering, and visualization in two or three dimensions.

When to use ICA. ICA is appropriate when the observations are believed to be mixtures of distinct latent processes and the goal is to separate or identify those processes. Typical examples include separating simultaneous voices or instruments, removing eye-blink and muscle artifacts from EEG, isolating physiological signals, and blind separation in telecommunications. ICA requires stronger assumptions than PCA, and its components are not ranked by importance.

The methods can also be complementary: PCA may first reduce dimension and whiten the observations, after which ICA estimates an additional rotation using non-Gaussian, higher-order information. Thus PCA is primarily a representation and compression method, whereas ICA is primarily a source-separation method.

5 Applications of ICA

ICA is useful whenever observations contain additive contributions from latent sources:

- **Audio processing:** separation of simultaneously recorded voices or instruments, speech enhancement, and microphone-array analysis.
- **Neuroscience:** removal of eye-blink, muscle, and line-noise artifacts from EEG or MEG, and exploratory decomposition of brain activity [6].
- **Biomedical signals:** isolation of physiological components and artifacts in ECG, imaging, and sensor recordings.
- **Image analysis:** feature extraction and decomposition of image collections into latent spatial patterns.
- **Telecommunications:** blind channel identification and source separation in multi-antenna systems.
- **Finance:** exploratory extraction of statistically independent factors from correlated asset returns, with careful attention to nonstationarity and interpretability.

The assumptions must always be checked against the application. Reverberant audio, propagation delays, nonlinear sensors, and time-varying mixtures require extensions beyond instantaneous linear ICA.

6 Reproducible audio demonstration

6.1 Input data and preprocessing

The repository contains three precomputed mono WAV mixtures: `ICA mix 1.wav`, `ICA mix 2.wav`, and `ICA mix 3.wav`. Each is a 16-bit PCM signal sampled at 44,100 Hz. The shortest common segment contains 264,515 samples, corresponding to 5.998 seconds. The $264{,}515 \times 3$ observation matrix has numerical rank three.

The notebook converts integer samples to floating point values, aligns all observations to the same length, and centers each channel. Three genuinely distinct observations are essential: duplicating one recording would produce a rank-deficient matrix and would not create enough information to recover three components.

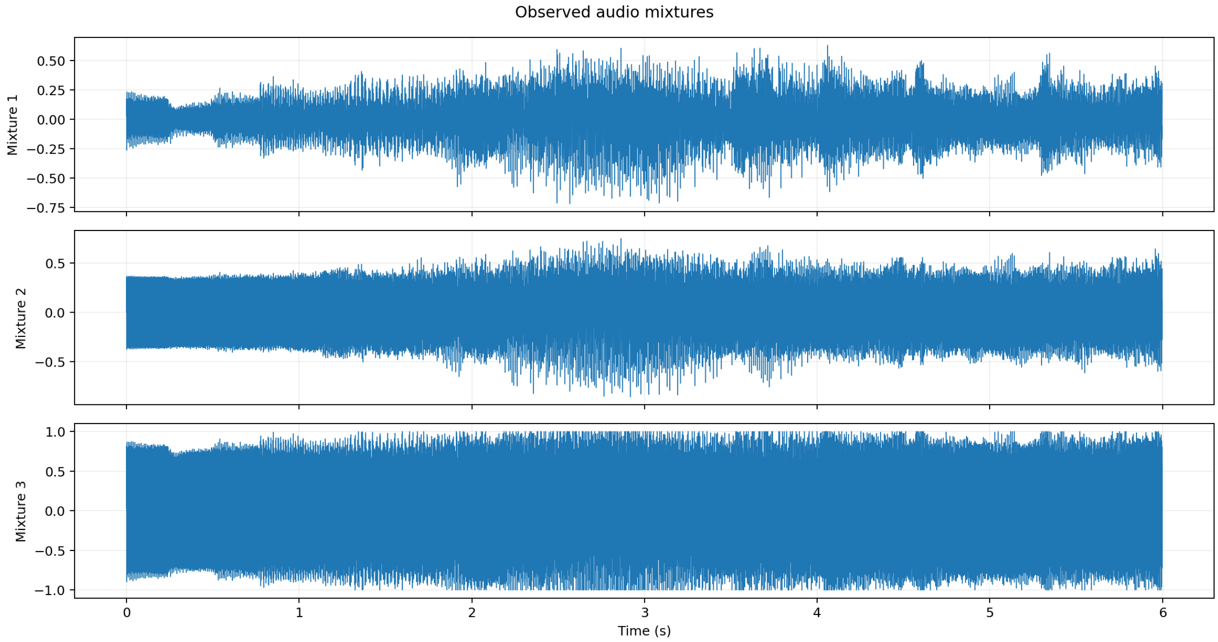

Figure 3: Observed audio mixtures used by the demonstration: three mono signals sampled at 44.1 kHz over 5.998 seconds.

6.2 FastICA configuration

The implementation uses scikit-learn's `FastICA` estimator with the parameters in Table 2 [7]. The random state is fixed for reproducibility. Unit-variance whitening is specified explicitly, and the `logcosh` contrast provides a smooth negentropy approximation.

Table 2: FastICA configuration used in the notebook.

Parameter	Value
Number of components	3
Algorithm	Parallel
Whitening	Unit variance
Nonlinearity	`logcosh`
Maximum iterations	2,000
Tolerance	10^{-5}
Random state	42

After estimation, each component is normalized independently to 95% of the signed 16-bit range before WAV export. This prevents clipping and integer overflow; it replaces the arbitrary volume multiplier used in the historical notebook.

7 Experimental results

7.1 Dependence in the observations

The observed mixtures are strongly correlated:

$$\mathbf{R}_X = \begin{bmatrix} 1.000 & 0.846 & 0.784 \\ 0.846 & 1.000 & 0.972 \\ 0.784 & 0.972 & 1.000 \end{bmatrix}. \tag{18}$$

The large off-diagonal values show that the observations share substantial linear structure.

7.2 Estimated independent components

FastICA converged in four iterations. The estimated components have an identity sample correlation matrix to the displayed precision. Their Fisher kurtosis values are

$$[1.401, -1.499, 2.775], \tag{19}$$

compared with $[-1.273, 1.290, 0.931]$ for the whitened PCA scores. The change in higher-order statistics illustrates the additional rotation performed by ICA after PCA whitening.

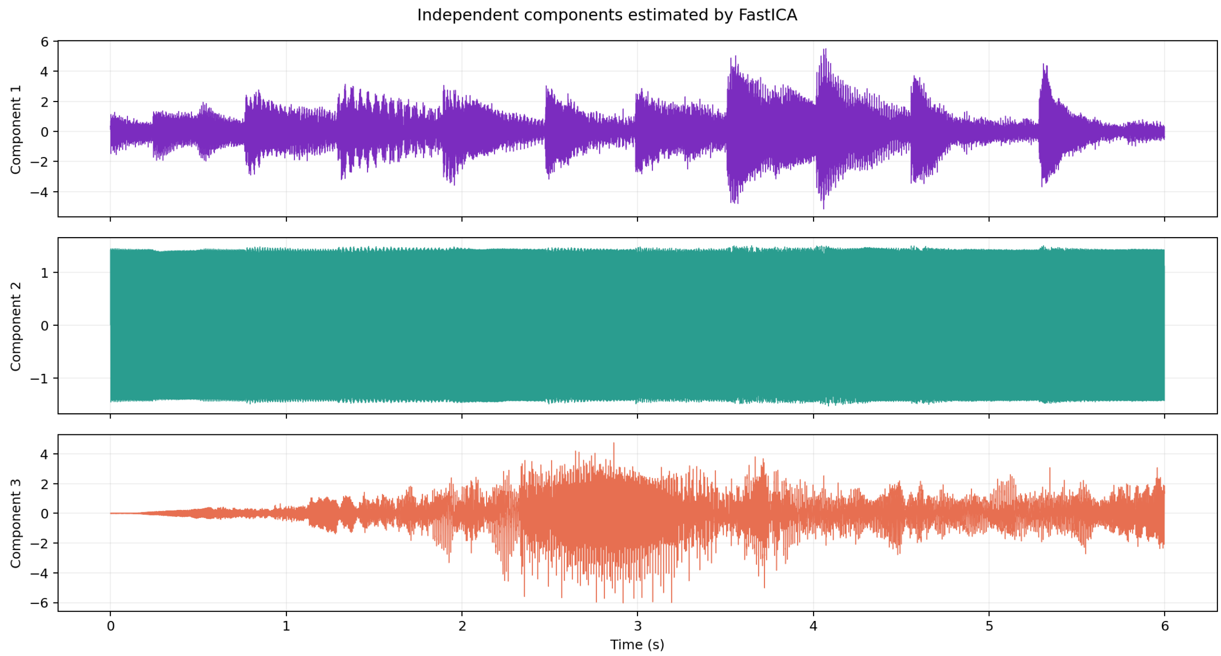

Figure 4: Three components estimated by FastICA. Their order, sign, and amplitude are not directly comparable with unknown source labels because of ICA's intrinsic ambiguities.

7.3 Numerical validation

The near-machine-precision reconstruction error confirms that the estimated mixing and unmixing transforms consistently factorize the same observation matrix. It is not a source-separation score: inverse transformation reconstructs the mixtures used for fitting.

The exported WAV files are stored in `outputs/separated/` and can be played inside the notebook. Listening provides a qualitative check, but a rigorous SDR or SI-SDR evaluation would require the isolated reference sources.

Table 3: Measured outputs of the reproducible experiment.

Metric	Result
FastICA iterations	4
Largest absolute off-diagonal ICA correlation	$< 10^{-4}$
Reconstruction RMSE	1.608×10^{-16}
Relative reconstruction RMSE	4.219×10^{-16}
Exported separated WAV files	3

8 Limitations and responsible interpretation

The experiment supports numerical convergence, decorrelation, non-Gaussian component estimation, reconstruction consistency, and qualitative listening. It does not support a claim of objectively perfect recovery because:

- the isolated reference sources and original mixing matrix are unavailable;
- zero correlation is weaker than statistical independence;
- the instantaneous model does not explicitly represent echo, propagation delay, nonlinear mixing, or a time-varying room;
- the method requires sufficiently distinct observations and a full-rank mixture;
- multiple Gaussian sources are not identifiable without additional structure.

These qualifications correct the stronger wording in the first report version while preserving its central theoretical motivation.

9 Conclusion

ICA is a fundamental method for discovering statistically independent structure in multivariate data. Its usefulness extends from the cocktail party problem to neuroscience, biomedical sensing, imaging, telecommunications, and exploratory factor analysis. The method relies on a linear mixture model, independence, non-Gaussianity, and sufficient observations.

PCA and ICA should not be treated as interchangeable. PCA identifies variance-preserving orthogonal directions and is preferred for dimensionality reduction, compact representation, and visualization. ICA exploits higher-order information to seek statistically independent latent processes and is preferred for blind source separation. PCA can also supply the dimension-reduction and whitening steps used before ICA.

The complete reproducible implementation, input mixtures, and separated outputs are available at

github.com/loukili27/ICA-Voices.

The code is intentionally kept in the version-controlled notebook rather than duplicated in a PDF appendix.

References

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